



SMOTE–IPF: Addressing the noisy and borderline examples problem in imbalanced classification by a re-sampling method with filtering

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ABSTRACT

Classification datasets often have an unequal class distribution among their examples. This problem is known as imbalanced classification. The *Synthetic Minority Over-sampling Technique* (SMOTE) is one of the most well-know data pre-processing methods to cope with it and to balance the different number of examples of each class. However, as recent works claim, class imbalance is not a problem in itself and performance degradation is also associated with other factors related to the distribution of the data. One of these is the presence of noisy and borderline examples, the latter lying in the areas surrounding class boundaries. Certain intrinsic limitations of SMOTE can aggravate the problem produced by these types of examples and current generalizations of SMOTE are not correctly adapted to their treatment.

This paper proposes the extension of SMOTE through a new element, an iterative ensemble-based noise filter called *Iterative-Partitioning Filter* (IPF), which can overcome the problems produced by noisy and borderline examples in imbalanced datasets. This extension results in SMOTE–IPF. The properties of this proposal are discussed in a comprehensive experimental study. It is compared against a basic SMOTE and its most well-known generalizations. The experiments are carried out both on a set of synthetic datasets with different levels of noise and shapes of borderline examples as well as real-world datasets. Furthermore, the impact of introducing additional different types and levels of noise into these real-world data is studied. The results show that the new proposal performs better than existing SMOTE generalizations for all these different scenarios. The analysis of these results also helps to identify the characteristics of IPF which differentiate it from other filtering approaches.

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1. Introduction

Several real-world classification problems in fields such as text categorization [49], medicine [52], bankruptcy prediction [38] and intrusion detection [31], are characterized by a highly imbalanced distribution of examples among the classes. In these problems, one class (known as the minority or positive class) contains a much smaller number of examples than the other classes (the majority or negative classes). The minority class is often the most interesting from the application point of

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view [11,5]. Class imbalance constitutes a difficulty for most learning algorithms which assume an approximately balanced class distribution and are biased toward the learning and recognition of the majority class. As a result, minority class examples usually tend to be misclassified.

The problem of learning from imbalanced data has been intensively researched in the last decade and several methods have been proposed to address it – for a review see, e.g., [24]. Re-sampling methods [9,8,33,47] are a classifier-independent type of techniques that modify the data distribution taking into account local characteristics of examples to change the balance between classes. There are numerous works discussing their advantages [4,10]. Among these methods, the *Synthetic Minority Over-sampling Technique* (SMOTE) [9] is one of the most well-known; it generates new artificial minority class examples by interpolating among several minority class examples that lie together.

However, some researchers have shown that the class imbalance ratio is not a problem itself. Even though the observation of a low classification performance in some concrete imbalanced problems may be influenced by the validation scheme used to estimate this performance of the classifiers [35], the classification performance degradation is usually linked to other factors related to data distributions [28,22,40]. Among them, in [40] the influence of noisy and borderline examples on classification performance in imbalanced datasets is experimentally studied. Borderline examples are defined as examples located either very close to the decision boundary between minority and majority classes or located in the area surrounding class boundaries where classes overlap. The authors of [33,40] refer to noisy examples as those from one class located deep inside the region of the other class. Furthermore, this paper, considers noisy examples in the wider sense of [57,43], in which they are treated as examples corrupted either in the attribute values or the class label.

Even though SMOTE achieves a better distribution of the number of examples in each class, when used in isolation it may obtain results that are not as good as they could be or it may even be counterproductive in many cases. This is because SMOTE presents several drawbacks related to its *blind* oversampling, whereby the creation of new positive (minority) examples only takes into account the closeness among positive examples and the number of examples of each class, whereas other characteristics of the data are ignored – such as the distribution of examples from the majority classes. These drawbacks, which can further aggravate the difficulties produced for noisy and borderline examples in the learning process, include: (i) the creation of too many examples around unnecessary positive examples which do not facilitate the learning of the minority class, (ii) the introduction of noisy positive examples in areas belonging to the majority class and (iii) the disruption of the boundaries between the classes and, therefore, an increase in the overlapping between them. In order to overcome these problems, two different approaches are followed in the literature:

1. Modifications of SMOTE (hereafter called *change-direction* methods). These guide the creation of positive examples performed by SMOTE towards specific parts of the input space, taking into account specific characteristics of the data. Within this group, the *Safe-Levels-SMOTE* (SL-SMOTE) [8], the *Borderline-SMOTE* (B1-SMOTE and B2-SMOTE) [23] or LN-SMOTE [37] methods are found, which try to create positive examples close to areas with a high concentration of positive examples or only inside the boundaries of the positive class.
2. Extensions of SMOTE by integrating it with additional techniques (these extensions will be referred to as *filtering-based* methods since SMOTE is integrated with either special *cleaning* or filtering methods). In the standard classification tasks, noise filters are often used in order to detect and eliminate noisy examples from training datasets and also to clean up and to create more regular class boundaries [55,53]. Experimental studies, such as [4], confirm the usefulness of integrating such filters – e.g., *Edited Nearest Neighbor Rule* (ENN) or *Tomek Links* (TL) [53] – as a post-processing step after using SMOTE.

The ability to deal with imbalanced datasets with noisy and borderline examples of methods belonging to both approaches will be studied in the experimental section, even though this paper also proposes a new extension of SMOTE. Existing extensions of SMOTE are very simple because they are based on using a single learning algorithm or simple measures such as, e.g., *k-Nearest Neighbors* (*k*-NN) [39] paradigm inside ENN [55] – used in SMOTE-ENN.

Some works highlight the good behavior of ensembles for classification in noisy environments, showing that the combined use of several classifiers is a good alternative in these scenarios as opposed to the employment of single classifiers [42,43]. In the same way, some authors also propose the usage of ensembles for filtering [7,17,18,54]. However, all these works only consider the point of view of the standard classification and the overall classification accuracy. Thus, ensembles are used for filtering in [7] considering that some examples have been mislabeled and the label errors are independent of particular classifiers learned from the data. In this scenario, the authors claim that collecting predictions from different classifiers could provide a better estimation of mislabeled examples rather than collecting information from a single classifier only. According to our best knowledge, these ensemble-based filters have not yet been used in the context of learning from imbalanced data. Analyzing such filters focuses our attention on the *Iterative-Partitioning Filter* (IPF) [32]. Its characteristics differentiate it from most of the filters, making it particularly suitable to overcome the problems produced by noisy and borderline examples specific to the dataset plus those additional ones that SMOTE may introduce.

The main aim of this paper is to propose and examine a new extension of SMOTE, in which the IPF noise filter is applied in post-processing resulting in SMOTE-IPF – its implementation can be found in KEEL¹ [2]. Its suitability for handling noisy and

¹ www.keel.es.

borderline examples in imbalanced data will be a particular focus of evaluation as these are one of the main sources of difficulties for learning algorithms. Differences between this approach and other re-sampling methods also based on generalizations of SMOTE will be discussed and studied. One cannot treat this proposal as a simple combination of two methods, as we want to study more deeply the conditions of its appropriate use in dealing with different types of noise in imbalanced data which have not been considered yet. We discuss its properties in comparison to other previous, related generalizations of SMOTE.

The other contribution of this paper is to provide a comprehensive experimental comparison of SMOTE–IPF with these generalizations. Moreover, different data factors will be considered in these parts of this experimental study. A first part will be carried out with special synthetic datasets containing different shapes of the minority class example boundaries and levels of borderline examples, as considered in related studies [22,28,29,40]. Additionally, a set of real-world datasets which are also known to be affected by noisy and borderline examples will be considered. All of these were used in [40] and are available in the KEEL-dataset repository [1]. Yet another contribution of this paper will be to introduce additional class or attribute noise into these real-world datasets and to study its impact on compared SMOTE generalizations. After preprocessing these datasets, the performances of the classifiers built with C4.5 [41] will be evaluated and they will also be contrasted using the proper statistical tests as recommended in the specialized literature [14,19,25]. The characteristics of IPF which differentiate it from other filters and a discussion on the strengths and weaknesses of IPF in dealing with imbalanced datasets with noisy and borderline examples will be analyzed in Section 6.

In addition, experiments with many other classification algorithms on the preprocessed datasets will be carried out in order to show the behavior of the preprocessing techniques with different classifiers. These are k -NN [39], a *Support Vector Machine* (SVM) [13,51], *Repeated Incremental Pruning to Produce Error Reduction* (RIPPER) [12] and PART [16]. Due to length restrictions, their results are only included on the web-page associated with this paper, available at <http://sci2s.ugr.es/noisebor-imbalanced>. This web-page also includes the basic information of this paper, the datasets used and the parameter setup for all the classification algorithms.

The rest of this paper is organized as follows. Section 2 presents the imbalanced dataset problem. Section 3 is devoted to the motivations behind our extension of SMOTE. Next, Section 4 describes the experimental framework. Section 5 includes the analysis of the experimental results, and Section 6 outlines the results and the suitability of IPF for the problem treated. Finally, in Section 7, some concluding remarks are presented.

2. Classification for imbalanced datasets

In this section, first the problem of imbalanced datasets is introduced in Section 2.1. Some additional problems related to class imbalance that may harm classifier performance are described in Section 2.2.

2.1. The problem of imbalanced datasets

The main difficulty of imbalanced datasets is that a standard classifier might ignore the importance of the minority class because its representation inside the dataset is not strong enough and the classifier is biased toward the majority class or, in other words, it is oriented to achieve a good total classification accuracy. Consequently, the examples that belong to the minority class are misclassified more often than those belonging to the majority class [27].

This type of data may be categorized depending on its imbalance ratio (IR) [15], which is defined as the relation between the majority class and minority class examples, by the expression

$$IR = \frac{N^-}{N^+} \quad (1)$$

where N^- and N^+ are the number of examples belonging to the majority and minority classes, respectively. Thus, a dataset is imbalanced when $IR > 1$.

A large number of approaches have been previously proposed to deal with the class imbalance problem. These approaches can be mainly categorized in two groups [3]:

1. *Algorithmic level approaches*. This group of methods tries to change search techniques or the classification decision strategies to impose bias toward the minority class or to improve the prediction performance by adjusting weights for each class [27].
2. *Data level approaches*. This group of methods preprocess the dataset modifying the data distribution to change the balance between classes considering local characteristics of examples [4].

Furthermore, cost-sensitive learning solutions incorporating both the data and algorithmic level approaches assume higher misclassification costs with samples in the minority class and seek to minimize the high cost errors [50].

There are some data level approaches particularly adapted to the usage of a concrete classifier. For example, the authors of [36] propose an evolutionary framework that uses an instance generation technique that modifies the original training set based on the performance of a specific classifier on the minority class. However, this is not the most common scenario and the great advantage of data level approaches is that they are more versatile, since their use is independent of the classifier

selected. Furthermore, one may preprocess all datasets beforehand in order to use them to train different classifiers. In this manner, the computation time needed to prepare the data is only required once. For these reasons, the proposal made in this paper belongs to this group of methods. In addition, re-sampling approaches can be categorized into two sub-categories: under-sampling [53,55], which consists of reducing the data by eliminating examples belonging to the majority class with the objective of balancing the number of examples of each class; and over-sampling [9,8], which aims to replicate or generate new positive examples in order to gain importance, improving the importance of this class.

In order to estimate the quality of classifiers built from imbalanced data, several measures have been proposed in the literature [24]. This is because the most widely used empirical measure, *total accuracy*, does not distinguish between the number of correct labels of different classes, which in the ambit of imbalanced problems may lead to erroneous conclusions. This paper considers the usage of the *Area Under the ROC Curve* (AUC) measure [6], which provides a single-number summary for the performance of learning algorithms and it is recommended in many other works in the literature [4,15].

2.2. Other factors characterizing imbalanced data

The imbalance ratio between classes is a problem that may hinder the performance of classifiers. However, it is not the only source of difficulty for classifiers; recent works have indicated other relevant issues related to the degradation of performance:

- *Presence of small disjuncts* [28,29] (Fig. 1a). The minority class can be decomposed into many sub-clusters with very few examples in each one, surrounded by majority class examples. This is a source of difficulty for most learning algorithms in detecting enough of these sub-concepts.
- *Overlapping between classes* [22,21] (Fig. 1b). There are often some examples from different classes with very similar characteristics, in particular if they are located in the regions around decision boundaries between classes. These examples refer to overlapping regions of classes.

Closely related to the overlapping between classes, in [40] another interesting problem in imbalanced domains is pointed out: the higher or lower presence of examples located in the area surrounding class boundaries, which are called *borderline examples*. Researchers have found that misclassification often occurs near class boundaries where overlapping usually occurs as well and it is hard to find a feasible solution for it [20]. The authors in [40] showed that classifier performance degradation was strongly affected by the quantity of *borderline examples* and that the presence of other noisy examples located farther outside the overlapping region also made the task of re-sampling methods very difficult.

This paper focuses on studying the influence of noisy and borderline examples in generalizations of SMOTE, considering the synthetic datasets and also the real-world ones used in [40] along with new noisy datasets built from the latter. These datasets will be described in Section 4. To clarify terminology, one must distinguish (inspired by [40,33]) between *safe*, *borderline* and *noisy examples* (see Fig. 2):

- *Safe examples* are placed in relatively homogeneous areas with respect to the class label.
- *Borderline examples* are located in the area surrounding class boundaries, where either the minority and majority classes overlap or these examples are very close to the difficult shape of the boundary – in this case, these examples are also difficult as a small amount of the attribute noise can move them to the wrong side of the decision boundary [33].
- *Noisy examples* are individuals from one class occurring in the safe areas of the other class. According to [33] they could be treated as examples affected by class label noise. Notice that the term *noisy examples* will be further used in this paper in the wider sense of [57], in which noisy examples are corrupted either in their attribute values or the class label.

The examples belonging to the last two groups often do not contribute to a correct class prediction [30]. Therefore, one could ask whether removing them (all or the most difficult misclassification parts) could improve classification performance.

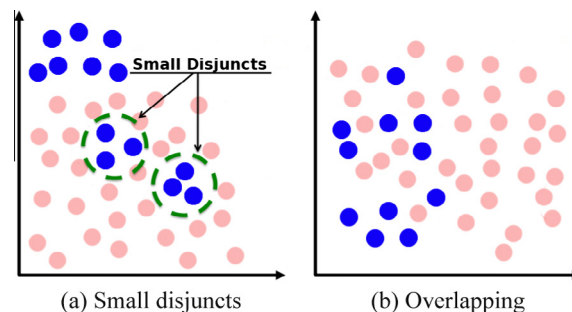


Fig. 1. Examples of the imbalance between classes: (a) small disjuncts and (b) overlapping between classes.

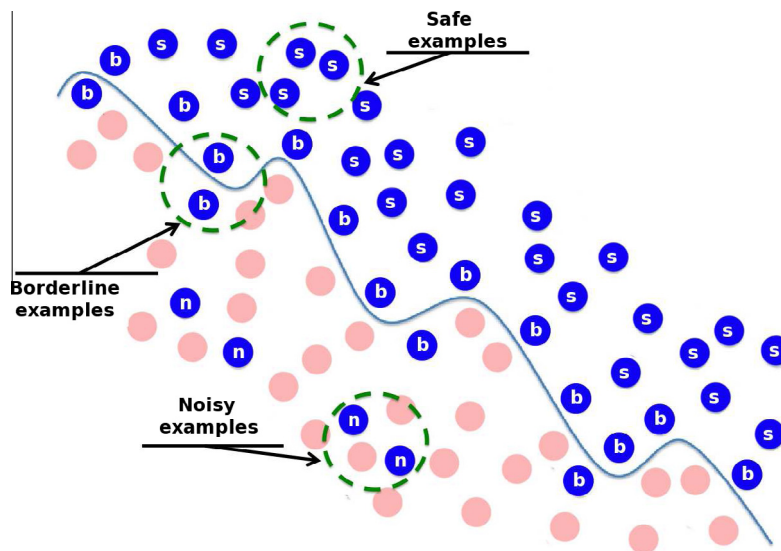


Fig. 2. The three types of examples considered in this paper: safe examples (labeled as *s*), borderline examples (labeled as *b*) and noisy examples (labeled as *n*). The continuous line shows the decision boundary between the two classes.

Thus, this paper proposes the use of noise filters to achieve this goal, because they are amply used with good results in classification. We are particularly interested in ensemble filters as they are the most careful when deciding whether an example should be viewed as noise and removed.

3. SMOTE along with noise filters to tackle noisy and borderline examples

In this section, first, the main details of the proposed extension of SMOTE are given in Section 3.1. Next, its two implicated parts are described in depth: the SMOTE algorithm in Section 3.2 and noise filters in Section 3.3.

3.1. Combining re-sampling and noise filters

As has been mentioned previously, SMOTE is one of the most well-known and widely used re-sampling techniques, however it may still have problems with some distributions of data. Two different generalizations of SMOTE have been proposed in the literature in order to improve final classification performance: *change-direction* and *filtering-based* methods – for the evaluation and discussion of their limitations see also [37]. The former, however, may present several important drawbacks when imbalanced datasets are suffering from noisy and borderline examples:

1. Noisy and borderline examples could be removed from the data to improve the final performance [30] but *change-direction* methods do not allow this option since their only function is to create new positive examples.
2. The creation of new positive examples, although directed towards specific parts of the input space, may be erroneous because it is based on data with noisy examples. This fact shows, once again, the need to introduce a cleaning phase after the creation of examples.

Methods based on extending SMOTE with additional cleaning seem to be more appropriate to deal with imbalanced problems with noisy and borderline examples. These methods follow two postulates:

1. Class distribution has to be transformed, e.g., balanced in some degree, in order to support the learning of classifiers [4].
2. The most difficult noisy and borderline examples should be removed from the training data since they are often the most difficult for learning [30,18].

The first task can be coped with by means of using re-sampling techniques. The SMOTE algorithm [9] helps to balance the class distribution, avoiding the overfitting problem that might be produced by using other techniques such as simple *Random Over-sampling* [4]. Furthermore, SMOTE can fill the interior of minority class sub-parts with synthetic minority class examples. This is a positive fact since in imbalanced datasets with a considerable quantity of borderline examples, minority class clusters are usually defined by those with an emptier interior. Nevertheless, class clusters may not be well defined in cases where some majority class examples might be invading the minority class space. The opposite can also be true, since

interpolating minority class examples can expand the minority class clusters, introducing artificial minority class examples too deeply into the majority class space. This additional minority noise is also caused by the blind over-generalization of SMOTE-based techniques of looking for nearest neighbors from the minority class only. Both situations could introduce additional noise into datasets.

The aforementioned problems are already well known. They have been tackled by combining SMOTE with an additional step of under-sampling, e.g., with the ENN filtering [55], which aims to remove mislabeled data from the training data after the usage of SMOTE. However, these methods do not perform this task as well as they should in all cases.

The second task requires specific and more powerful methods designed to eliminate mislabeled examples when datasets have a considerable number of such examples. A group of methods that address this problem is ensemble-based noise filters [18,7,54]. This paper proposes the extension of SMOTE with one of these filters: the IPF filter [32], which will be responsible for removing noisy examples originally present in the dataset and also those created by SMOTE. Besides this, IPF cleans up class boundaries, making them more regular and facilitating in this way the posterior learning phase [18].

These two techniques (SMOTE and IPF) must be applied to the imbalanced dataset in the correct order in order to obtain reasonable final results: SMOTE in the first place and then the IPF noise filter. This is due to IPF being designed to deal with standard classification datasets. Its application over an imbalanced dataset before the usage of SMOTE (which balances the distribution of classes) may carry the risk of removing all the examples from the minority class, which may be seen as noisy examples because they are underrepresented in the dataset. In short, as a summary and justification of this approach, it is claimed that:

1. The SMOTE algorithm fulfills a dual function: it balances the class distribution and it helps to fill in the interior of sub-parts of the minority class.
2. The IPF filter removes the noisy examples originally present in the dataset and also those created by SMOTE. Besides this, IPF cleans up the boundaries of the classes, making them more regular.

Note that the scheme proposed (SMOTE–IPF) enables one to replace SMOTE with any of its modifications, in particular, the change-direction generalizations. However, in this paper we prefer to use the basic version of SMOTE because it is consistent with earlier research on the usage of filtering techniques and facilitates the comparison with them. The usage of IPF may also disturb the effects of the earlier modifications of examples by the change-direction methods. Moreover, some of these methods strongly focus on the over-sampling of concrete regions of the original data. For example, *Borderline-SMOTE* may over-strength the boundary zone, which will be problematic for studying data with many noisy and borderline examples.

3.2. The synthetic minority over-sampling technique

SMOTE [9], introduced by Chawla and co-authors, is now one of the most popular over-sampling methods. In this approach, the positive class is over-sampled by taking each minority class example and introducing synthetic examples along the line segments joining any/all of the k minority class nearest neighbors. In order to find these neighbors in the space of numerical and nominal attributes, the HVDM metric is applied [56]. Depending on the amount of oversampling required, neighbors from the k nearest neighbors are randomly chosen. Analyzing the current literature on the usage of SMOTE, one can notice that $k = 5$ neighbors is usually chosen. This procedure of building a local neighborhood is also often applied in other resampling methods, such as SPIDER. Although one could ask a more general question by tuning a particular k value depending on the given data characteristics, we decided to use one value $k = 5$ to be more consistent with other related works on SMOTE and its generalizations since they were compared using the same data sets as we have chosen for our study. Taking the same motivations to be consistent with related works, we tune the oversampling amount to balance both classes to 50%.

Synthetic examples are generated in the following way. Take the difference between the feature vector (sample) under consideration and its nearest neighbor. Multiply this difference by a random number between 0 and 1, and add it to the feature vector under consideration. This causes the selection of a random point along the line segment between two specific features. This approach effectively forces the decision region of the minority class to become more general.

3.3. Noise filters

Noise filters are preprocessing mechanisms designed to detect and eliminate noisy examples in the training set [55,7,32,44]. The result of noise elimination in preprocessing is a reduced training set which is then used as an input to a machine learning algorithm.

Some of these filters are based on the computation of different measures over the training data. For instance, the method proposed in [18] is based on the observation that the elimination of noisy examples reduces the *Complexity of the Least Complex Correct Hypothesis* value of the training set.

In addition, there are many other noise filters based on the usage of ensembles. In [7], multiple classifiers belonging to different learning paradigms were built and trained from a possibly corrupted dataset and then used to identify mislabeled data, which is characterized as the examples that are incorrectly classified by the multiple classifiers. Similar techniques

have been widely developed considering the building of several classifiers with the same learning algorithm [17,54]. Instead of using multiple classifiers learned from the same training set, in [17] a *Classification Filter* (CF) approach is suggested, in which the training set is partitioned into n subsets, then a set of classifiers is trained from the union of any $n - 1$ subsets; those classifiers are used to classify the examples in the excluded subset, eliminating the examples that are incorrectly classified.

From our knowledge and preliminary, earlier experiments performed with different ensemble-based filters, e.g., the *Ensemble Filter* (EF) [7], the *Cross-Validated Committees Filter* (CVCF) [54], CF and others, the notable good behavior of the *Iterative-Partitioning Filter* [32] when detecting noisy examples must be pointed out. IPF has characteristics which differentiate it from most of the noise filters and may provide the reasons why it performs better than them – they will be analyzed and discussed in Section 6.

IPF removes noisy examples in multiple iterations until a stopping criterion is reached. The iterative process stops when, for a number of consecutive iterations k , the number of identified noisy examples in each of these iterations is less than a percentage p of the size of the original training dataset. Initially, the method starts with a set of noisy examples $A = \emptyset$. The basic steps of each iteration as follows:

1. Split the current training dataset E into n equal sized subsets.
2. Build a classifier with the C4.5 algorithm over each of these n subsets and use them to evaluate the whole current training dataset E .
3. Add to A the noisy examples identified in E using a voting scheme (consensus or majority).
4. Remove the noisy examples: $E \leftarrow E \setminus A$.

Two voting schemes can be used to identify noisy examples: consensus and majority. The former removes an example if it is misclassified by all the classifiers, whereas the latter removes an example if it is misclassified by more than half of the classifiers.

The parameter setup for the implementation of IPF used in this work has been determined experimentally in order to better fit it to the characteristics of imbalanced datasets with noisy and borderline examples once they have been preprocessed with SMOTE. More precisely, the majority scheme is used to identify the noisy examples, $n = 9$ partitions with random examples in each one are created and $k = 3$ iterations for the stop criterion and $p = 1\%$ of removed examples are considered. This parameter setup is based on a study of the influence of each parameter on the results – the justification of each parameter value and its influence is given in Section 6.3.

4. Experimental framework

In this section, the details of the experimental study developed in this paper are presented. First, in Section 4.1, we describe how the synthetic imbalanced datasets with borderline examples were built. Then, the real-world datasets and the noise introduction processes are presented in Section 4.2. In Section 4.3 the preprocessing techniques considered in this work are briefly described. Finally, in Section 4.4, the methodology of the analysis carried out is described.

4.1. Synthetic imbalanced datasets with borderline examples

This paper uses the family of synthetic datasets used in prior research on the role of borderline examples [40]. These data were created following other experimental studies of small disjuncts [28,29] and overlapping between classes [21]. However, these studies were focused on studying single factors and very simple (lines and rectangles) shapes of decision boundaries. Thus, the authors of [40] created more complex data affected by many factors, including borderline examples. Many datasets with different configurations were generated by special software and evaluated; for more details see [46]. In this paper we consider these configurations of datasets which were the basis for the previous analysis of the role of borderline examples for different basic classifiers, such as C4.5, and re-sampling methods [40]. These datasets are briefly characterized below:

1. *Number of classes and attributes.* This work focuses on binary classification problems (the minority versus the majority class) with examples randomly and uniformly distributed in the two-dimensional real-value space.
2. *Number of examples and imbalance ratios.* Multiple datasets with two different numbers of examples and imbalance ratios are considered: datasets with 600 examples and $IR = 5$ and datasets with 800 examples and $IR = 7$. The values of the parameters resulted from the assumption of having at least 20 examples for the subpart of the decomposed minority class. Smaller cardinalities led to unstable results [46].
3. *Shapes of the minority class.* Three different shapes of the minority class surrounded uniformly by the majority class are taken into account:
 - *Sub-cluster* (Fig. 3a): the examples from the minority class are located inside rectangles following related works on small disjuncts [28].
 - *Clover* (Fig. 3b): represents a more difficult, non-linear setting, in which the minority class resembles a flower with elliptic petals.

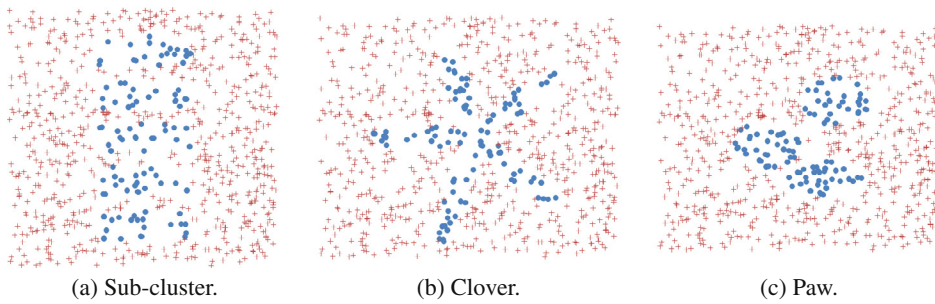


Fig. 3. Shapes of the minority class.

- *Paw* (Fig. 3c): the minority class is decomposed into 3 elliptic subregions of varying cardinalities, in which two subregions are located close to each other, and the smaller sub-region is separated.

The minority class is decomposed into 5 parts except for paw data with 3 subregions. Paw could better represent real-world data than clover. Moreover, both clover and paw should be more difficult to learn than the simple circles that were considered in some related works.

4. *Disturbance ratios (DR)*. The impact of disturbing the borders of sub-regions in the minority class will be studied. This is carried out by increasing the ratio of borderline examples, i.e., the disturbance ratio, from the minority class subregions. 5 levels of DR are considered: 0%, 30%, 50%, 60% and 70%. The width of the borderline overlapping areas is comparable to the width of the safe parts of sub-regions. The aforementioned DR values result from earlier studies [46] which showed that smaller values (less than 20%) did not affect the performance of the considered classifier very much. Moreover, higher values will better discriminate between the usefulness of different preprocessing methods [40].

In such a way, 30 synthetic datasets with the above mentioned properties have been managed, all of which are available on the web-page associated with this paper.

4.2. Real-world datasets

The choice of the real-world datasets was based on the work on imbalanced classification with noisy and borderline examples presented in [40]. These are available at the KEEL-dataset repository² [1] or have been provided by the authors of [45] in the case of the *acl* dataset. Multi-class datasets are modified to obtain two-class imbalanced problems, defining the joint of one or more classes as positive and the remainder as negative. Table 1 shows the characteristics of these datasets. For each one, the number of examples (#Exa), attributes (#Att), the name of the minority class (Min), the number of examples of the minority class (#Min), of the majority class (#Maj) and the imbalance ratio (IR) are shown.

These data are also characterized by a large number of difficult examples, which could be recognized as borderline or noisy ones. In [37] a simple analysis of a local neighborhood has been made with *k*-NN classifiers according to rules applied in *Borderline-SMOTE* [23]. So, an example was treated as noisy if all its neighbors belong to opposite classes, as a borderline example if it was misclassified by the majority of the opposite classes and safe if it was correctly classified. Nearly all of these datasets were difficult with respect to the minority class as they contained much less safe examples than others. In particular, in the *cleveland* dataset the minority class contains 35 examples with 22 noisy and 13 borderline ones. A similar situation occurs for *breast cancer*. Some other datasets, e.g., *haberman* or *ecoli*, contain more borderline examples than noisy ones (e.g. *haberman* has 51 and 20 respectively plus 10 safe ones). As the number of noisy examples is quite high compared to the size of the minority class these data sets are chosen to study their impact. The exception is *newthyroid*, which seems to contain easier to understand data as its number of safe examples is higher than the others.

Furthermore, we decided that these real-world datasets could be transformed into more complex and even more difficult versions with an artificially increased noise level. In such a way, two different noise levels will be introduced into them: $x = 20\%$ and $x = 40\%$. The introduction of noise consists of separately corrupting either the class labels or the attribute values of some examples belonging to each dataset. These corruptions respectively belong to the noise categories formally known as class noise and attribute noise and were amply used and studied in [57], a reference paper in the framework of noisy data in classification. The same two noise introduction schemes proposed by the authors of that paper are used in this work, so that:

- Class noise is introduced into the datasets following a pairwise scheme in which a majority class example has a probability of $x/100$ to be incorrectly labeled as belonging to the minority class.
- Attribute noise is introduced into the original datasets corrupting each attribute separately. To corrupt each attribute A_i , approximately $x\%$ of the examples in the dataset are chosen and the value of A_i of each of these examples is assigned a

² <http://www.keel.es/datasets.php>.

Table 1

Characteristics of the real-world datasets.

Dataset	#Exa	#Att	Min	#Min	#Maj	IR
acl	140	6	with knee injury	40	100	2.5
breast	286	9	recurrence-events	85	201	2.36
bupa	345	6	sick	145	200	1.38
cleveland	303	13	positive	35	268	7.66
ecoli	336	7	imU	35	301	8.60
haberman	306	3	died	81	225	2.78
hepatitis	155	19	die	32	123	3.84
newthyroid	215	5	hyper	35	180	5.14
pima	768	8	positive	268	500	1.87

random value between the minimum and maximum of the domain of that attribute following a uniform distribution (if A_i is numerical), or choosing a random value of the domain (if A_i is nominal).

The performance estimation of each classifier for each of these real-world datasets, and also for the synthetic ones, is obtained by means of 5 runs of a stratified 5-fold cross-validation and their results are averaged. Dividing the dataset into 5 folds is considered in order to dispose of a sufficient quantity of minority class examples in the test partitions. In this way, test partition examples are more representative of the underlying knowledge and meaningful performance results can be obtained.

4.3. Re-sampling techniques for comparison

Different re-sampling techniques to adjust the class distribution in the training data based on generalizations of SMOTE are studied in this paper. The usefulness of a new SMOTE extension is studied in a comprehensive comparative study with other, related versions of SMOTE, in particular the best known representations of *change-direction* and *filtering-based* methods. Table 2 shows the SMOTE-based methods considered in this study – a wider description of such methods is found on the web-page associated with this paper.

Note that SMOTE-TL and SMOTE-ENN are approaches based on extending SMOTE with an additional filtering (*filtering-based* methods), whereas SL-SMOTE, B1-SMOTE and B2-SMOTE are approaches based on directing the creation of the positive examples (*change-direction* methods).

4.4. Analysis methodology

In order to check the suitability of the proposed extension of SMOTE versus the other re-sampling techniques when dealing with imbalanced datasets with noisy and borderline examples, the experiments are divided into three differentiated parts depending on the type of datasets considered in each one: synthetic, real-world and noisy modified real-world datasets.

The effect of the aforementioned preprocessing techniques will be analyzed comparing the AUC for each dataset obtained with C4.5 [41], which has been used in many other works in imbalanced classification, in particular concerning SMOTE [47,48]. Furthermore, it is known to be more sensitive to different factors of imbalanced data than, e.g., SVM [13] and is often used inside the ensembles. The standard parameters along with a post-pruning have been considered for the executions.

For each of the three types of datasets, the AUC results obtained by C4.5 for our approach against (i) not applying preprocessing and applying SMOTE alone, (ii) the other *filtering-based* methods (SMOTE-ENN and SMOTE-TL) and (iii) the *change-direction* methods (B1-SMOTE, B2-SMOTE and SL-SMOTE) will be separately compared. We have separately studied the differences between our proposal and the filtering-based and change-direction methods for two main reasons. First, the separation is motivated by the different nature of the methods of both groups that share common characteristics, which allow us to independently obtain conclusions with each kind of method. On the other hand, performing a multiple statistical comparison usually requires a much higher quantity of datasets to detect significant differences when the number of comparison methods increases. Multiple statistical comparisons are then limited by the number of datasets, and the comparison

Table 2

Re-sampling techniques considered.

Method	Reference	Method	Reference
SMOTE	[9]	SL-SMOTE	[8]
SMOTE-TL	[4]	B1-SMOTE	[23]
SMOTE-ENN	[4]	B2-SMOTE	[23]

grouping the two types of methods (filtering-based and change-direction) can only be performed if a much higher quantity of datasets than the one considered in this paper is available for study.

Additionally, statistical comparisons in each of these cases will be also performed. Wilcoxon's signed ranks statistical test [14] will be applied to compare SMOTE-IPF with no preprocessing and the usage of SMOTE alone. This is a nonparametric pairwise test that aims to detect significant differences between two sample means; that is, the behavior of the two algorithms involved in each comparison. The results of the two methods involved in the comparison over all the datasets will be compared using Wilcoxon's test and the p -values associated with these comparisons will be obtained. The p -value represents the lowest level of significance of a hypothesis that results in a rejection and it allows one to know both whether two algorithms are significantly different and the degree of their difference.

Regarding the comparison between our approach and the other re-sampling techniques (either *filtering-based* or *change-direction* methods), the aligned Friedman's procedure [19] will be used. This is an advanced nonparametric test for performing multiple comparisons, which improves the classic Friedman test. The Friedman test is based on sets of ranks, one set for each data set; and the performances of the algorithms analyzed are ranked separately for each data set. Such a ranking scheme allows for intra-set comparisons only, since inter-set comparisons are not meaningful. When the number of algorithms for comparison is small, this may pose a disadvantage. In such cases, comparability among data sets is desirable and we can employ the method of aligned ranks [26]. Because of this, we will use the aligned Friedman's test to compute the set of ranks that represent the effectiveness associated with each algorithm and the p -value related to the significance of the differences found by this test. In addition, the adjusted p -value with Hochberg's test [25] will be computed. More information about these tests and other statistical procedures can be found at <http://sci2s.ugr.es/sicidm/>.

We will consider a difference to be significant if the p -value obtained is lower than 0.1 [14,19] – even though p -values slightly higher than 0.1 might be also showing important differences.

5. Evaluation of re-sampling methods with noisy and borderline examples

In this section, the performance of C4.5 using the different preprocessing techniques over the imbalanced datasets with noisy and borderline examples is analyzed. In Section 5.1, the results considering synthetic datasets are analyzed, whereas Sections 5.2 and 5.3 are respectively devoted to analyzing the results on the real-world datasets and the noisy modified real-world ones.

5.1. Results on synthetic datasets

Table 3 presents the AUC results obtained by C4.5 on each synthetic dataset when preprocessing with each re-sampling approach considered in this paper. The column denoted by *None* corresponds to the case in which no re-sampling is performed prior to C4.5. The best case for each dataset is highlighted in bold. From these results, the following main points should be stressed:

- Increasing *DR*, fixing a shape of the minority class and an *IR*, strongly deteriorates the performance of C4.5 without preprocessing.
- Preprocessing improves the performance with respect to the case without preprocessing in nearly all the datasets. The improvements in the results for each single dataset reflect this fact.
- SMOTE-IPF obtains better results than the rest of the re-sampling methods in 11 of the 30 datasets considered and obtains results close to the best performances in the rest of the cases.
- Without preprocessing, linear rectangle shapes (sub-cluster datasets) are easier to learn than non-linear ones (clover or paw), since the former obtain higher performances. However, with most of the preprocessing techniques the non-linear paw datasets even outperform the linear sub-cluster datasets at the same *DR* level.
- The highest improvements of SMOTE-IPF are obtained in the learning of non-linear datasets, since 8 of the 11 overall best performance results are obtained for these types of datasets, which are the most difficult ones.

Table 4 collects the results of applying Wilcoxon's signed ranks statistical test between SMOTE-IPF versus *None* and SMOTE. As the p -values ($p_{Wilcoxon}$) and the sums of ranks (R^+ and R^-) reflect, the application of SMOTE-IPF produces an improvement in the results obtained with respect to not preprocessing or preprocessing only with SMOTE with these synthetic imbalanced datasets with borderline examples.

Regarding the comparison between re-sampling techniques considering the synthetic datasets, Table 5 presents the ranks of the aligned Friedman's procedure (Rank column) for each group of techniques (*filtering-based* methods and *change-direction* ones). In the case of the Friedman's aligned rank test, the method with the best average ranking among all the datasets is considered to be the best. Please note that SMOTE-IPF is established in all the cases as the control algorithm because it has obtained the best aligned Friedman's rank, indicating the high performance of the approach.

The p -value related to the significance of the differences found by the aligned Friedman's test ($p_{AlignedFriedman}$ row) is also shown. In addition, the $p_{Hochberg}$ column shows the adjusted p -value with Hochberg's test. Post-hoc tests indicate those methods that the control algorithm outperforms significantly.

Table 3

AUC results obtained by C4.5 on synthetic datasets with borderline examples.

Dataset	None	SMOTE	SMOTE-ENN	SMOTE-TL	SL-SMOTE	B1-SMOTE	B2-SMOTE	SMOTE-IPF
sub – cluster _{IR=5, DR=0}	94.10	93.00	90.00	91.30	91.50	94.20	93.50	92.70
sub – cluster _{IR=5, DR=30}	64.80	81.90	81.80	81.30	82.20	79.20	80.00	83.40
sub – cluster _{IR=5, DR=50}	51.30	80.80	77.80	81.00	80.40	77.30	77.50	77.80
sub – cluster _{IR=5, DR=60}	52.00	78.60	77.00	77.80	78.10	77.00	73.10	79.70
sub – cluster _{IR=5, DR=70}	50.00	77.50	77.00	79.20	82.30	78.50	78.00	80.10
sub – cluster _{IR=7, DR=0}	87.50	94.79	93.07	92.29	90.79	95.42	95.64	95.21
sub – cluster _{IR=7, DR=30}	76.86	81.57	79.64	77.64	81.71	80.29	79.71	83.00
sub – cluster _{IR=7, DR=50}	53.86	78.29	75.29	78.50	81.29	75.79	74.79	78.57
sub – cluster _{IR=7, DR=60}	50.00	80.21	75.71	78.57	81.36	73.93	73.86	79.79
sub – cluster _{IR=7, DR=70}	50.00	82.50	78.21	81.07	81.21	76.43	73.14	80.93
clover _{IR=5, DR=0}	70.50	83.50	86.80	87.30	85.70	86.40	86.80	85.50
clover _{IR=5, DR=30}	54.30	83.80	83.40	84.20	81.10	81.20	81.90	85.30
clover _{IR=5, DR=50}	51.60	83.40	81.00	84.40	83.00	79.90	83.10	82.40
clover _{IR=5, DR=60}	56.50	80.80	80.50	81.50	78.70	76.90	78.70	82.20
clover _{IR=5, DR=70}	50.00	77.20	76.10	79.40	77.10	74.10	78.70	79.00
clover _{IR=7, DR=0}	70.71	87.93	87.79	89.36	88.21	87.14	88.93	91.64
clover _{IR=7, DR=30}	53.21	83.64	83.14	81.00	84.36	79.29	81.29	85.71
clover _{IR=7, DR=50}	50.00	81.21	82.86	82.00	78.71	78.93	73.36	81.93
clover _{IR=7, DR=60}	50.00	78.79	77.71	82.21	77.21	77.79	71.71	82.14
clover _{IR=7, DR=70}	51.57	78.29	76.07	78.86	78.29	75.57	72.79	80.21
paw _{IR=5, DR=0}	91.00	96.30	94.90	94.90	92.80	93.10	93.30	94.50
paw _{IR=5, DR=30}	70.00	84.40	86.40	84.20	84.50	84.80	87.30	84.90
paw _{IR=5, DR=50}	67.90	84.60	83.30	83.90	85.00	81.70	82.80	84.90
paw _{IR=5, DR=60}	54.10	81.00	81.30	81.90	82.30	76.30	78.90	83.30
paw _{IR=5, DR=70}	57.70	82.80	83.50	84.40	82.70	80.60	80.40	83.40
paw _{IR=7, DR=0}	68.29	93.43	93.93	93.43	92.93	95.21	93.79	94.29
paw _{IR=7, DR=30}	56.71	84.29	84.93	84.21	83.50	84.36	85.43	85.29
paw _{IR=7, DR=50}	50.00	85.00	84.79	84.86	84.29	78.14	79.71	85.79
paw _{IR=7, DR=60}	53.00	83.36	80.71	82.57	83.00	72.71	75.14	82.14
paw _{IR=7, DR=70}	50.00	82.57	80.57	79.93	84.14	78.64	77.71	85.71

The best case for each dataset is highlighted in bold.

Table 4Wilcoxon's test results for the comparison of SMOTE-IPF (R^+) versus None and SMOTE (R^-) on synthetic datasets considering the AUC results obtained by C4.5.

Methods	R^+	R^-	P_{Wilcoxon}
SMOTE-IPF vs. None	464.0	1.0	<0.0001
SMOTE-IPF vs. SMOTE	366.0	99.0	0.0050

Table 5

Multiple statistical comparison with synthetic datasets.

	Algorithm	Rank	P_{Hochberg}
Filtering	SMOTE-IPF	26.73	–
	SMOTE-ENN	64.93	< 0.000001
	SMOTE-TL	44.83	0.007289
	$P_{\text{AlignedFriedman}}$	0.000014363186	
Change-Dir.	SMOTE-IPF	30.30	–
	SL-SMOTE	47.85	0.050698
	B1-SMOTE	83.15	<0.000001
	B2-SMOTE	80.70	<0.000001
	$P_{\text{AlignedFriedman}}$	0.00002425464	

Looking at Table 5, one can make some observations:

- The average rank for SMOTE-IPF obtained by the aligned Friedman's test is the best, i.e. the lowest, in all the considered cases and it is notably differentiated from the ranks of the rest of the methods. This occurs with both *filtering-based* and *change-direction* methods.
- Comparing ranks for *filtering-based* methods, SMOTE-IPF is followed by SMOTE-TL. SMOTE-ENN obtains the highest rank notably differentiated from the rest.

- Among the *change-direction* methods, SL-SMOTE outperforms the ranks of B1-SMOTE and B2-SMOTE, which are quite similar.
- The *p*-values of the aligned Friedman's test are very low in all the scenarios, which shows a great significance in the differences found.
- The adjusted *p*-values by Hochberg's test are very low in all comparisons.

From the results of Tables 3–5, one can conclude that SMOTE-IPF performs better than other SMOTE versions when dealing with the synthetic imbalanced datasets built with borderline examples, particularly in those with non-linear shapes of the minority class. All the statistical tests also clearly show the statistical significance of this better performance of SMOTE-IPF.

5.2. Results on real-world datasets

Table 6 presents the AUC results obtained by C4.5 on each real-world dataset with noisy and borderline examples when preprocessing with each re-sampling approach. With these types of datasets, the application of preprocessing techniques does not always obtain the expected results. For instance, some of the selected preprocessing methods, such as SMOTE-ENN, SMOTE-TL and B2-SMOTE, do not outperform the performance of *None* on *acl* and *bupa* datasets. *acl* could be considered as an easier dataset for basic methods (this has also been observed in [40]) while *bupa* could be characterized by other complex data factors apart from the presence of noisy and borderline examples.

Wilcoxon's test between SMOTE-IPF and *None* and the aligned Friedman's and Hochberg's multiple comparison tests comparing SMOTE-IPF along with the rest of re-sampling techniques are respectively found in Tables 7 and 8.

From the AUC results of Table 6 and the statistical comparisons performed in Tables 7 and 8, one should note that:

- The Wilcoxon's test between SMOTE-IPF and *None*, and between SMOTE-IPF and SMOTE again clearly shows an improvement in the results obtained with respect to not preprocessing or applying SMOTE alone.

Table 6

AUC results obtained by C4.5 on real-world datasets with noisy and borderline examples.

Dataset	None	SMOTE	SMOTE-ENN	SMOTE-TL	SL-SMOTE	B1-SMOTE	B2-SMOTE	SMOTE-IPF
acl	88.75	86.75	86.75	88.00	85.25	89.00	88.00	88.50
breast	61.73	60.56	63.70	62.01	64.72	63.31	63.58	64.40
bupa	64.40	66.88	61.46	60.18	66.84	68.60	63.61	67.53
cleveland	52.58	54.85	57.22	64.33	60.07	54.75	56.66	62.82
ecoli	72.46	82.16	89.97	82.33	84.52	79.55	79.37	86.55
haberman	57.57	65.41	64.68	62.03	67.07	61.40	60.23	66.76
hepatitis	67.66	71.38	71.90	71.15	68.53	66.39	62.70	72.25
newthyroid	90.87	96.35	94.64	94.37	90.95	93.45	97.18	96.63
pima	70.12	71.29	71.40	69.48	73.97	70.94	73.77	73.58

The best case for each dataset is highlighted in bold.

Table 7

Wilcoxon's test results for the comparison of SMOTE-IPF (R^+) versus *None* and SMOTE (R^-) on real-world datasets considering the AUC results obtained by C4.5.

Methods	R^+	R^-	P_{Wilcoxon}
SMOTE-IPF vs. <i>None</i>	44.0	1.0	0.0078
SMOTE-IPF vs. SMOTE	45.0	0.0	0.0039

Table 8

Multiple statistical comparison with real-world datasets.

	Algorithm	Rank	P_{Hochberg}
Filtering	SMOTE-IPF	6.89	–
	SMOTE-ENN	16.00	0.014890
	SMOTE-TL	19.11	0.002178
	$P_{\text{AlignedFriedman}}$	0.037679463832	
Change-Dir.	SMOTE-IPF	9.33	–
	SL-SMOTE	16.11	0.172352
	B1-SMOTE	23.67	0.007804
	B2-SMOTE	24.89	0.005208
	$P_{\text{AlignedFriedman}}$	0.064566807525	

- SMOTE-IPF only obtains the best results on one single dataset (*hepatitis*) considering all the preprocessing methods. However, the Friedman's rank of SMOTE-IPF is clearly the best result compared with the rest of re-sampling techniques if the performance of all the datasets is summarized. This shows the great robustness of SMOTE-IPF when applied to the real-world datasets.
- The p -values of the aligned Friedman's test are very low in all the scenarios, which shows a great significance in the differences found.
- The adjusted p -values by Hochberg's test are also very low in all comparisons, particularly with the *filtering-based* methods, even though with SL-SMOTE the p -value obtained is a little higher. Therefore, the suitability of SMOTE-IPF to address imbalanced real-world datasets with noisy and borderline examples is statistically shown.

Comparing the results of the synthetic datasets in Table 5 with respect to those of the real-world ones in Table 8, one observes that some of the methods that obtain the better Friedman's ranks on synthetic datasets, such as SMOTE-TL among the *filtering-based* methods, obtain less notable results on real-world ones. The latter data are perhaps more complex than the former and they may require more elaborate techniques. SMOTE-IPF remains the best method considering its aligned Friedman's rank and the Hochberg's test p -values with both synthetic and real-world datasets.

5.3. Results on noisy modified real-world datasets

This section is devoted to the analysis of the AUC results obtained by C4.5 on the noisy modified real-world datasets (see Table 9). From this table, one can observe that:

- Outperforming the results of *None* by the re-sampling methods is often more difficult when dealing with more noisy real-world datasets than with real-world datasets without additional noise. For instance, the performance of *None* on the

Table 9

AUC results obtained by C4.5 on noisy modified real-world datasets with both class noise and attribute noise.

	Dataset	None	SMOTE	SMOTE-ENN	SMOTE-TL	SL-SMOTE	B1-SMOTE	B2-SMOTE	SMOTE-IPF
Class Noise	acl _{x=20%}	86.50	87.25	81.75	82.00	85.50	83.75	81.75	88.25
	breast _{x=20%}	63.32	63.44	62.96	59.78	64.44	61.31	67.12	64.15
	bupa _{x=20%}	60.75	63.35	56.05	63.47	62.89	61.94	64.81	61.46
	cleveland _{x=20%}	66.07	61.47	50.00	70.93	60.33	59.79	58.83	63.51
	ecoli _{x=20%}	73.58	75.86	83.55	75.45	78.38	75.64	79.37	78.42
	haberman _{x=20%}	62.18	62.61	60.95	53.97	62.65	59.82	61.03	62.25
	hepatitis _{x=20%}	70.37	62.09	66.90	71.68	69.87	62.10	60.89	77.58
	newthyroid _{x=20%}	92.06	87.98	90.44	79.40	89.92	84.17	89.72	88.21
	pima _{x=20%}	68.99	72.70	71.63	66.09	67.28	69.69	70.09	73.08
	acl _{x=40%}	68.50	71.00	74.00	62.50	66.75	66.75	73.50	74.75
	breast _{x=40%}	56.41	57.26	61.87	53.65	55.54	57.28	59.52	55.86
	bupa _{x=40%}	55.03	52.10	55.29	50.68	53.42	55.53	56.59	56.02
	cleveland _{x=40%}	57.02	58.97	61.41	59.57	52.65	59.19	59.94	62.98
	ecoli _{x=40%}	60.76	55.65	69.61	53.36	56.48	61.10	61.89	71.12
	haberman _{x=40%}	50.00	50.00	53.16	51.19	50.00	50.00	50.00	61.92
	hepatitis _{x=40%}	50.00	53.60	56.49	58.57	61.79	56.67	50.00	65.54
	newthyroid _{x=40%}	50.00	51.03	57.58	53.53	51.03	56.59	51.03	53.29
	pima _{x=40%}	62.63	60.75	54.19	53.40	61.72	61.56	62.16	61.85
Attribute noise	acl _{x=20%}	73.25	68.25	73.00	68.25	70.75	72.75	74.00	70.00
	breast _{x=20%}	54.56	64.03	63.40	55.47	60.77	61.01	59.46	62.81
	bupa _{x=20%}	53.10	51.47	56.23	51.31	55.35	52.36	55.23	58.41
	cleveland _{x=20%}	53.33	57.30	56.54	58.62	57.34	54.28	54.66	63.89
	ecoli _{x=20%}	59.93	71.70	64.81	62.65	74.03	65.61	67.48	72.89
	haberman _{x=20%}	55.68	59.09	63.25	57.42	58.07	60.42	59.41	62.81
	hepatitis _{x=20%}	78.26	65.10	72.69	70.99	75.58	74.52	85.23	78.74
	newthyroid _{x=20%}	80.60	87.26	83.61	83.02	85.32	86.90	84.05	87.58
	pima _{x=20%}	66.71	65.85	67.64	63.72	63.99	63.17	64.80	69.99
	acl _{x=40%}	83.25	79.50	79.75	80.75	84.00	82.50	79.50	86.25
	breast _{x=40%}	52.00	55.52	60.14	50.99	56.51	57.28	53.08	56.53
	bupa _{x=40%}	57.40	61.28	58.98	54.97	59.88	61.66	57.54	57.93
	cleveland _{x=40%}	59.99	50.00	58.74	62.73	64.61	50.94	61.52	65.69
	ecoli _{x=40%}	65.81	70.78	74.13	74.56	72.14	70.46	67.61	73.02
	haberman _{x=40%}	50.18	62.88	63.23	61.21	61.30	62.33	58.60	66.22
	hepatitis _{x=40%}	76.61	63.34	65.08	61.39	70.78	69.29	70.06	77.34
	newthyroid _{x=40%}	83.77	89.52	88.13	90.16	87.58	88.93	90.04	91.55
	pima _{x=40%}	62.87	63.84	65.48	65.42	68.47	64.26	63.99	66.91

The best case for each dataset is highlighted in bold.

newthyroid _{$\chi=20\%$} and *pima* _{$\chi=40\%$} datasets with class noise are the best results with respect to considering the use of preprocessing.

- The observation of the best results in each single dataset show that SMOTE–IPF is the best method in 8 of 18 class noise datasets, whereas it is the best in 9 of 18 attribute noise datasets. SMOTE–ENN and B2–SMOTE are also notable, with each obtaining 3 of 18 of the best results in class noise datasets and 2 of 18 in attribute noise datasets.

Table 10 shows the results of comparing the application of Wilcoxon's test to SMOTE–IPF with *None* and SMOTE, whereas Table 11 shows the aligned Friedman's test and the Hochberg's test results comparing SMOTE–IPF with respect to the rest of re-sampling techniques.

From these tables, one can make some observations:

- The need to apply advanced preprocessing techniques is shown by the low p -values obtained in Table 10, even though a slightly higher p -value (0.1551) is obtained comparing SMOTE–IPF and *None* with 20% of class noise.
- SMOTE–IPF is established with both types of noise (class and attribute noise), both noise levels (20% and 40%) and with both types of techniques (*filtering-based* and *change-direction* methods) as the control algorithm because it obtains the best aligned Friedman's rank.
- The p -values of the aligned Friedman's test are low with both class and attribute noise. However, the most significant differences are found when comparing against the *filtering-based* techniques rather than against the *change-direction* ones.
- The adjusted p -values by Hochberg's test are generally low in all comparisons, with the exception of 20% of class noise in the *change-direction* group of methods, in which the highest p -values are obtained. The p -values are lower in the case of the attribute noise datasets than in the class noise datasets for a noise level of 20%. However, increasing the noise level up to 40% makes the differences found more significant independently of the type of noise.

Therefore, SMOTE–IPF also performs well in this scenario with these datasets, with additional noise induced as the AUC results and statistical comparisons show. The better performance and statistical significance of the results of SMOTE–IPF with the attribute noise datasets must be pointed out. This property seems to be important due to the fact that attribute noise is much more common than class noise in real-world datasets, as indicated in [57].

5.4. Analysis of the noise filters in imbalanced datasets preprocessed with SMOTE

Since the only difference among all the existing *filtering-based* methods is the type of noise filter used, in this section the results of the filtering process are analyzed. Four additional criteria will be studied based on the percentage of examples removed by the noise filters after preprocessing with SMOTE. We analyze the percentage of examples filtered with respect to the set of: (i) all the examples (%Total), (ii) the synthetic positive examples created by SMOTE (%Synt), (iii) the original positive examples (%Pos) and (iv) the original negative examples (%Neg). For the sake of simplicity, only ENN and IPF will be compared and also only the noise level (20%) will be studied, even though the results of the other filtering method (TL) and the other noise level (40%) are found in the web-page associated with this paper.

Table 10

Wilcoxon's test results for the comparison of SMOTE–IPF (R^+) versus *None* and SMOTE (R^-) on real-world datasets with both class noise and attribute noise considering the AUC results obtained by C4.5.

Methods	Class noise – 20%			Class noise – 40%			Attribute noise – 20%			Attribute noise – 40%		
	R^+	R^-	p_{Wilcoxon}	R^+	R^-	p_{Wilcoxon}	R^+	R^-	p_{Wilcoxon}	R^+	R^-	p_{Wilcoxon}
SMOTE–IPF vs. <i>None</i>	34.0	11.0	0.1551	42.0	3.0	0.0195	43.0	2.0	0.0117	45.0	0.0	0.0039
SMOTE–IPF vs. SMOTE	37.0	8.0	0.0977	43.0	2.0	0.0117	42.0	3.0	0.0195	39.0	6.0	0.0546

Table 11

Multiple statistical comparison with real-world datasets with both class noise and attribute noise.

Algorithm	Class noise – 20%		Class noise – 40%		Attribute noise – 20%		Attribute noise – 40%	
	Rank	p_{Hochberg}	Rank	p_{Hochberg}	Rank	p_{Hochberg}	Rank	p_{Hochberg}
SMOTE–IPF	9.22	–	7.56	–	6.11	–	7.44	–
SMOTE–ENN	15.56	0.090521	12.89	0.154044	13.67	0.043455	16.00	0.022221
SMOTE–TL	17.22	0.065019	21.56	0.000366	22.22	0.000033	18.56	0.005964
$p_{\text{AlignedFriedman}}$	0.03280041552		0.040217689132		0.043181723863		0.037237166607	
SMOTE–IPF	11.00	–	9.11	–	9.11	–	8.89	–
SL–SMOTE	15.56	0.359013	26.11	0.001859	20.33	0.025274	16.00	0.152201
B1–SMOTE	28.44	0.001332	20.39	0.046325	24.33	0.006531	21.33	0.024445
B2–SMOTE	19.00	0.214458	18.39	0.061755	20.22	0.025274	27.78	0.000428
$p_{\text{AlignedFriedman}}$	0.065317774636		0.067460983239		0.06469719834		0.069232529882	

Table 12 shows the percentage of examples removed in synthetic datasets with borderline examples. The analysis of these results leads to the following observations:

- ENN usually removes more examples (%Total) in *sub-cluster* and *paw* datasets, whereas IPF removes more examples only in *clover* datasets. One must note that *clover* datasets have more non-linear shapes of the minority class than the rest of the types of datasets.
- The %Synt results show similar conclusions to those of %Total results. It is important to point out that ENN does not remove any examples in *clover* datasets.
- ENN usually removes large quantities of examples of the original positive examples (%Pos), whereas IPF removes very low quantities independently of the type of dataset.
- IPF usually removes more examples of the negative class (%Neg) in *sub-cluster* and *paw* datasets, whereas it removes less examples in *clover* datasets.

Table 13 shows the percentage of examples removed in real-world datasets with noisy and borderline examples. These results show the following points:

1. **Real-world datasets.** IPF removes more synthetic positive examples (%Synt) and more original negative examples (%Neg) in almost all the datasets, where as it always removes less original positive examples (%Pos).
2. **Noisy modified real-world datasets.** IPF removes more synthetic examples (%Synt) in almost all the datasets and it removes less original positive examples (%Pos) with both types of noise. However, there is an important difference in the percentage of negative examples removed %Neg in both types of noise: IPF generally removes less negative examples than ENN with class noise datasets and more examples with attribute noise datasets.

The percentages of the total number of examples removed %Total do not provide significant results in any of the aforementioned cases. Furthermore, it is important to remark that, in many datasets, ENN does not remove any synthetic examples.

From the results shown in this section, it can be observed that an important characteristic of the filtering performed by IPF is that it usually removes less original positive examples than ENN. One must not forget that, although noisy examples

Table 12

Percentages of examples removed by ENN and IPF after preprocessing the synthetic datasets with SMOTE.

Dataset	ENN %Total	IPF	ENN %Synt	IPF	ENN %Pos	IPF	ENN %Neg	IPF
sub – cluster _{IR=5, DR=0}	14.30	7.65	9.00	6.25	19.00	0.00	17.60	10.30
sub – cluster _{IR=5, DR=30}	17.83	14.65	10.19	9.44	41.50	3.00	19.20	21.15
sub – cluster _{IR=5, DR=50}	20.88	17.00	11.44	6.69	50.00	1.75	22.60	28.30
sub – cluster _{IR=5, DR=60}	18.23	16.72	9.31	8.63	49.50	1.50	19.10	26.25
sub – cluster _{IR=5, DR=70}	19.90	17.93	10.69	7.50	49.00	1.25	21.45	29.60
sub – cluster _{IR=7, DR=0}	12.73	4.64	6.96	7.08	22.75	0.00	16.25	3.21
sub – cluster _{IR=7, DR=30}	15.68	14.23	8.67	5.42	45.50	2.00	17.43	23.54
sub – cluster _{IR=7, DR=50}	17.27	15.52	9.08	6.83	54.50	2.50	18.96	24.82
sub – cluster _{IR=7, DR=60}	16.11	17.45	9.25	5.42	54.50	0.25	16.50	30.21
sub – cluster _{IR=7, DR=70}	16.89	18.29	7.92	5.42	57.75	0.25	18.75	31.89
clover _{IR=5, DR=0}	3.05	8.80	0.00	2.19	13.00	1.25	3.50	15.60
clover _{IR=5, DR=30}	6.75	11.83	0.00	3.94	37.25	0.50	6.05	20.40
clover _{IR=5, DR=50}	9.05	13.75	0.00	3.81	53.00	1.50	7.50	24.15
clover _{IR=5, DR=60}	10.18	13.10	0.00	2.88	60.50	0.25	8.25	23.85
clover _{IR=5, DR=70}	10.87	14.93	0.00	3.69	65.50	0.75	8.65	26.75
clover _{IR=7, DR=0}	2.64	6.05	0.00	1.54	19.75	1.00	2.46	10.64
clover _{IR=7, DR=30}	4.77	10.84	0.00	3.04	44.25	0.00	3.21	19.07
clover _{IR=7, DR=50}	6.98	11.75	0.00	2.75	63.50	1.50	4.89	20.93
clover _{IR=7, DR=60}	8.13	12.00	0.00	2.62	71.25	0.50	6.07	21.68
clover _{IR=7, DR=70}	8.09	12.70	0.00	2.88	75.25	0.50	5.43	22.86
paw _{IR=5, DR=0}	7.75	4.08	2.06	1.25	7.50	0.25	12.35	7.10
paw _{IR=5, DR=30}	14.48	9.25	7.50	1.94	20.75	1.25	18.80	16.70
paw _{IR=5, DR=50}	17.20	9.58	9.00	2.75	23.50	0.25	22.50	16.90
paw _{IR=5, DR=60}	16.25	10.85	5.06	2.31	28.00	1.25	22.85	19.60
paw _{IR=5, DR=70}	17.38	10.47	7.56	1.69	30.75	1.00	22.55	19.40
paw _{IR=7, DR=0}	7.45	4.02	2.75	1.50	7.75	0.25	11.43	6.71
paw _{IR=7, DR=30}	13.21	8.34	5.71	2.54	22.25	0.50	18.36	14.43
paw _{IR=7, DR=50}	14.98	8.98	6.38	2.92	26.00	0.75	20.79	15.36
paw _{IR=7, DR=60}	14.46	10.93	4.63	2.42	32.25	1.00	20.36	19.64
paw _{IR=7, DR=70}	15.50	10.68	5.79	2.17	36.50	0.50	20.82	19.43

The best case for each dataset and criteria is highlighted in bold.

Table 13

Percentages of examples removed by ENN and IPF after preprocessing the real-world and noisy modified real-world datasets with SMOTE.

	Dataset	ENN %Total	IPF	ENN %Synt	IPF	ENN %Pos	IPF	ENN %Neg	IPF
Real-world	acl	7.50	15.00	0.42	22.50	26.88	7.50	4.00	13.50
	breast	37.06	29.97	38.50	30.72	40.42	31.48	34.82	28.91
	bupa	33.81	23.50	0.00	22.73	46.55	27.07	33.88	21.13
	cleveland	8.92	9.26	0.00	3.42	87.86	29.29	6.11	11.65
	ecoli	5.73	7.85	0.00	4.51	48.57	2.14	5.82	11.46
	haberman	20.44	27.17	13.02	24.98	52.46	21.32	13.67	30.67
	hepatitis	10.19	17.18	0.00	18.01	49.67	1.67	10.73	19.51
	newthyroid	1.46	4.44	0.00	8.28	9.29	1.43	1.11	1.94
Class Noise	pima	21.10	20.67	0.65	17.89	42.82	19.31	18.95	22.70
	acl _{x=20%}	22.58	20.88	4.22	32.05	38.82	30.63	16.50	10.70
	breast _{x=20%}	42.62	35.42	34.52	38.25	47.09	46.70	40.98	25.88
	bupa _{x=20%}	41.74	33.69	0.00	24.17	48.10	39.56	38.96	28.64
	cleveland _{x=20%}	24.54	22.04	0.00	12.45	75.96	51.47	19.14	16.24
	ecoli _{x=20%}	19.09	25.37	0.00	26.48	60.74	29.16	15.88	23.22
	haberman _{x=20%}	35.39	31.90	14.22	28.94	50.20	33.52	32.03	31.70
	hepatitis _{x=20%}	19.77	31.22	0.00	34.18	51.80	30.74	15.91	30.01
Attribute Noise	newthyroid _{x=20%}	16.69	21.93	0.00	34.52	49.28	36.22	10.10	8.57
	pima _{x=20%}	33.83	26.77	2.52	25.85	40.32	26.40	31.97	27.21
	acl _{x=20%}	18.00	19.88	2.50	23.75	54.37	13.75	12.75	20.00
	breast _{x=20%}	35.59	31.09	37.81	34.75	31.16	34.88	36.10	27.37
	bupa _{x=20%}	40.44	32.25	0.00	23.64	58.79	26.38	38.25	38.88
	cleveland _{x=20%}	8.25	10.49	0.00	4.73	85.71	32.14	5.06	12.59
	ecoli _{x=20%}	8.72	11.09	0.94	4.98	52.14	3.57	10.55	17.36
	haberman _{x=20%}	21.56	26.94	10.77	25.52	52.77	18.83	17.22	30.78
	hepatitis _{x=20%}	11.05	21.48	0.00	24.21	60.67	4.00	10.20	22.71
	newthyroid _{x=20%}	5.00	9.58	0.00	14.66	40.00	5.00	2.22	6.39
	pima _{x=20%}	29.23	25.95	6.25	23.06	52.42	23.32	27.45	28.70

The best case for each dataset and criteria is highlighted in bold.

exist in the original dataset, the minority class is usually under-represented and the classifier should mainly reflect the properties of these original examples. Thus, being more conservative with regard to removing too many examples from the minority class seems to be an advantage.

IPF removes more synthetic positive examples created by SMOTE than ENN in real-world and noisy modified real-world datasets, along with the more complex non-linear synthetic datasets (*clover*). These types of datasets are the most complex considered in this paper and it is therefore more likely that the synthetic examples introduced by SMOTE will be noisy.

With noisy real-world datasets – which are more likely to suffer from attribute noise than from class noise [57] – and noisy modified real-world datasets with attribute noise, the number of original negative examples removed by IPF is larger than with other noise filters, whereas with class noise datasets the quantity of negative examples removed by IPF is generally fewer. This is due to real-world and attribute noise datasets having larger quantities of negative examples suffering from noise than class noise datasets. It seems to be logical to delete more negative examples in these types of datasets than in class noise datasets and this is indeed the behavior of IPF. Therefore, this fact leads one to think that IPF performs a more accurate filtering than other noise filters in these scenarios.

6. SMOTE–IPF: suitability of the approach, strengths and weaknesses

This section summarizes the main conclusions obtained in the experimental section. Section 6.1 outlines the results obtained with the different types of datasets. Then, Section 6.2 describes the characteristics of IPF that make it suitable for this type of problem when preprocessed with SMOTE. Section 6.3 analyzes the main drawback of SMOTE–IPF, its parametrization. Finally, Section 6.4 establishes a hypothesis in order to explain its good behavior in the different scenarios considered.

6.1. Results obtained with the different types of datasets

The experimental results shows that SMOTE–IPF statistically outperforms the rest of the methods with all the types of datasets considered: synthetic datasets with borderline examples, real-world datasets with noisy and borderline examples and noisy modified real-world datasets. It is particularly suitable for the most complex types of problems: the non-linear synthetic datasets and the noisy modified real-world problems. Within the latter group, it particularly stands out with attribute noise datasets, which is the most common type of noise [57] and, in this case, it is also the most disruptive one due it

affecting both classes, whereas class noise only affects the majority class. The increase in the noise level makes the differences in favor of SMOTE–IPF still more remarkable.

6.2. Characteristics of IPF and suitability for problems preprocessed with SMOTE

One important fact that makes IPF suitable for imbalanced datasets with noisy and borderline examples preprocessed with SMOTE is its *iterative elimination of noisy examples*. This fact implies that the examples removed in one iteration do not influence detection in subsequent ones, resulting in a more accurate noise filtering.

Furthermore, the *ensemble nature* of IPF enables it to collect predictions from different classifiers which may provide a better estimation of difficult noisy examples, as apposed to collecting information from a single classifier only [7]. IPF also enables the *creation of more different classifiers* – using random partitions, for example – than other ensemble-based filters due to it providing more freedom when creating the partitions from which these classifiers are built. Creating diversity among the classifiers built is a key factor when ensembles of classifiers are used [34]. Finally, unlike other ensemble-based filters such as EF, which uses different classification algorithms to build the classifiers, IPF *only requires one classification algorithm* and is thus simpler.

6.3. On the parametrization of IPF

The choice of the different parameters of IPF can be seen as its main drawback, since there are numerous parameters and the behavior of the filter is quite dependent on their values. From our many experiments we can draw several conclusions regarding the influence of the different parameters on the performance results.

We have confirmed that using the majority scheme leads to better results than the consensus scheme since the number of noisy and borderline examples is large enough in comparison with the quantity of safe examples. The consensus scheme is very strict in removing examples and does not enable one to remove enough examples to change the performance significantly.

Regarding the number of partitions, a larger number usually implies better noise detection (and also a higher preprocessing time) since the voting scheme depends on more information. It is recommended that this number be odd, in order to avoid ties in the votes of the classifiers. Considering $n = 9$ partitions leads to a good balance between computational cost and performance.

The way to build the partitions enables one to control the diversity among classifiers. We have tested different strategies to create the partitions, such as stratified cross-validation – e.g. EF or CVCF – or random partitions. Random partitions were considered because they lead to better performance results since, as was expected, the partitions and, therefore the classifiers built, are more different.

The rest of the parameters allow a wider range of possibilities obtaining similar performances. Standard parameters recommended by the authors of IPF also work well with our SMOTE-preprocessed imbalanced datasets, so they are fixed to $k = 3$ iterations for the stop criterion and $p = 1\%$ for the percentage of removed examples.

6.4. Hypothesis to explain the good behavior of IPF with respect to other filters

The properties of IPF seem to be well adapted to the removal of noisy and borderline examples, implying an advantage over other noise filters. Most of the noise filters combined with SMOTE, such as ENN or TL, have a noise identification based on distances among examples to their nearest neighbors, taking into account their classes. This issue, although overlooked in the literature, may be an important drawback: since SMOTE is based on the distance to the nearest neighbors to create a new positive example, such synthetic examples introduced by SMOTE, although noisy, are highly likely to not be identified by filters based on distances to the nearest neighbors. Using a noise identification method based on more complex rules, such as IPF, enables one to group larger quantities of examples with similar characteristics, although exceptions exist that will be considered to be noise, avoiding the aforementioned problem and detecting noisy examples easily.

7. Concluding remarks

This paper has focused on the presence of noisy and borderline examples, which is an important and contemporary research issue for learning classifiers from imbalanced data. It has been proposed to extend SMOTE with a new element, the IPF noise filter, to control the noise introduced by the balancing between classes produced by SMOTE and to make the class boundaries more regular. The suitability of the approach in this scenario has been analyzed.

Synthetic imbalanced datasets with different shapes of the minority class, imbalance ratios and levels of borderline examples have been considered. A set of real-world imbalanced datasets with different quantities of noisy and borderline examples and other factors have been also considered. Additional noise has been introduced into the latter considering two noise schemes: a class noise scheme and an attribute noise scheme. All these datasets have been preprocessed with our proposal and several re-sampling techniques that can be found in the literature. Finally, the C4.5 algorithm has been tested over these preprocessed datasets.

The values of the AUC measure have shown that our proposal has a notably better performance when dealing with imbalanced datasets with noisy and borderline examples with both synthetic and real-world datasets. SMOTE–IPF also outperforms the rest of the methods with the real-world datasets with additional noise. Our proposal especially outperforms the rest of the methods with the more complex to learn datasets in each group of datasets: the non-linear synthetic datasets and the attribute noise real-world datasets. These observations are supported by statistical tests.

The experiments performed with others classifiers (k -NN, SVM, RIPPER and PART) have provided similar results and conclusions on the superiority of SMOTE–IPF to those shown for C4.5. Although statistical differences are less significant using SVM and RIPPER with attribute noise datasets, SMOTE–IPF obtains the better performances and the aligned Friedman's ranks. The especially good results of SMOTE–IPF on the synthetic datasets using RIPPER and on the class noise datasets using k -NN must also be pointed out. These results have been better than those of C4.5 shown here.

One must consider that the ensemble-nature of IPF, which constitutes a robust and accurate way of detecting mislabeled examples, the iterative noise detection and elimination processes carried out and the possibility of controlling the diversity between classifiers are the key points of IPF which finally produce a more accurate filtering process. All these factors help SMOTE–IPF to obtain better performances than other re-sampling techniques in the scenarios considered.

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